

[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 24 September 2026, 10:24 AM**State** Finished**Completed on** Thursday, 24 September 2026, 10:34 AM**Time taken** 9 mins 31 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int arr[n];
6      for(int i=0;i<n;i++){
7          scanf("%d",&arr[i]);
8      }
9      for(int i=0;i<n;i++){
10         for(int j=i+1;j<n;j++){
11             if(arr[i]==arr[j]){
12                 printf("%d",arr[i]);
13                 return 0;
14             }
15         }
16     }
17 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 24 September 2026, 10:34 AM**State** Finished**Completed on** Thursday, 24 September 2026, 10:58 AM**Time taken** 23 mins 25 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     int slow = a[0];
13     int fast = a[0];
14
15     // Find intersection point
16     do {
17         slow = a[slow];
18         fast = a[a[fast]];
19     } while (slow != fast);
20
21     // Find duplicate
22     slow = a[0];
23
24     while (slow != fast) {
25         slow = a[slow];
26         fast = a[fast];
27     }
28
29     printf("%d", slow);
30
31     return 0;
32 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 24 September 2026, 10:58 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:12 AM**Time taken** 13 mins 57 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include<stdio.h>
2
3 int main() {
4     int t;
5     if (scanf("%d", &t) != 1) return 0;
6     while (t--) {
7         int n;
8         scanf("%d", &n);
9         int arr[n];
10        for(int i=0; i<n; i++) {
11            scanf("%d", &arr[i]);
12        }
13
14        int m;
15        scanf("%d", &m);
16        int brr[m];
17        for(int i=0; i<m; i++) {
18            scanf("%d", &brr[i]);
19        }
20        for(int i=0; i<n; i++) {
21            for(int j=0; j<m; j++) {
22                if(arr[i] == brr[j]) {

```

```
23         printf("%d ", arr[i]);
24         break;
25     }
26 }
27 }
28 printf("\n");
29 }
30 return 0;
31 }
32 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 24 September 2026, 11:13 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:19 AM**Time taken** 5 mins 46 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int t;
5     if (scanf("%d", &t) != 1) return 0;
6
7     while (t--) {
8         int n;
9         if (scanf("%d", &n) != 1) break;
10        int arr[n];
11        for (int i = 0; i < n; i++) {
12            scanf("%d", &arr[i]);
13        }
14        int m;
15        if (scanf("%d", &m) != 1) break;
16        int brr[m];
17        for (int i = 0; i < m; i++) {
18            scanf("%d", &brr[i]);
19        }
20
21        int i = 0, j = 0;
22        int first = 1;

```

```

23 while (i < n && j < m) {
24     if (arr[i] < brr[j]) {
25         i++;
26     } else if (arr[i] > brr[j]) {
27         j++;
28     } else {
29         if (!first) {
30             printf(" ");
31         }
32         printf("%d", arr[i]);
33         first = 0;
34
35         i++;
36         j++;
37     }
38 }
39 printf("\n");
40 }
41
42 return 0;
43 }
44

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 24 September 2026, 11:19 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:26 AM**Time taken** 6 mins 38 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      if (scanf("%d", &n) != 1) return 0;
6
7      int a[n];
8      for (int i = 0; i < n; i++) {
9          scanf("%d", &a[i]);
10     }
11
12     int k;
13     if (scanf("%d", &k) != 1) return 0;
14
15     for (int i = 0; i < n; i++) {
16         for (int j = i + 1; j < n; j++) {
17             if (a[j] - a[i] == k) {
18                 printf("1");
19                 return 0;
20             }
21         }
22     }
23
24     printf("0");
25     return 0;
26 }
27

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓

	Input	Expected	Got	
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 24 September 2026, 11:27 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:34 AM**Time taken** 7 mins 25 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n,k;
4      scanf("%d",&n);
5      int a[n];
6      for(int i=0;i<n;i++){
7          scanf("%d",&a[i]);
8      }
9      scanf("%d",&k);
10     int i=0,j=1;
11     while(i<n && j<n){
12         if(i==j){
13             j++;
14         }
15         else if(a[j]-a[i]<k){
16             j++;
17         }
18         else if(a[j]-a[i]>k){
19             i++;
20         }
21         else{
22             printf("1");
23             return 0;
24         }
25     }
26     printf("0");
27     return 0;
28
29 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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